THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET C

Chapters 8 & 9: **Introduction to Trigonometry, Some Applications of Trigonometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures for questions on heights and distances. Take $\sqrt{3} = 1.732$ where needed.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. $\sin 0^\circ + \cos 0^\circ$ equals: 1

(a) $\frac{1}{2}$ **(b)** 2 **(c)** 0 **(d)** 1

2. If $\tan\theta = \sqrt{3}$ (θ acute), then θ is: 1

(a) 30° **(b)** 45° **(c)** 90° **(d)** 60°

3. $2\sin 30^\circ \cos 30^\circ$ equals: 1

(a) $\frac{\sqrt{3}}{2}$ **(b)** $\frac{1}{2}$ **(c)** 1 **(d)** $\sqrt{3}$

4. A student states: "$\cos A$ can exceed 1 for some acute angle A." The correct response is: 1

(a) Correct near 0° **(b)** Correct

(c) Incorrect — base < hypotenuse **(d)** Cannot be determined

5. For $1 + \cot^2\theta$, Student A says $\csc^2\theta$; Student B says $\sec^2\theta$. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

6. The shadow of a 20 m pole when the sun's elevation is 60° is: 1

(a) 20 m **(b)** 10 m **(c)** $\frac{20}{\sqrt{3}}$ m **(d)** $20\sqrt{3}$ m

7. The angle of elevation of the top of a 10 m wall from a point $10\sqrt{3}$ m away is: 1

(a) 30° **(b)** 60° **(c)** 45° **(d)** 90°

8. From a point 50 m from its foot, the angle of elevation of a tree top is 45°. The height of the tree is: 1

(a) 50 m **(b)** $\frac{50}{\sqrt{3}}$ m **(c)** 25 m **(d)** $50\sqrt{3}$ m

9. A student states: "As the sun rises higher, shadows become longer." The correct response is: 1

(a) Incorrect — shadows become shorter **(b)** Correct

(c) No change **(d)** Cannot be determined

10. An aeroplane at a height of 3000 m is seen at 60° elevation from A. Student A says its horizontal distance is $1000\sqrt{3}$ m; Student B says $3000\sqrt{3}$ m. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. In $\triangle ABC$ right-angled at B, $AB = 5$ cm and $AC = 13$ cm. Find $\sin A$ and $\tan C$. **2**

12. Evaluate: $\frac{\sin^2 45^\circ + \cos^2 45^\circ}{\tan^2 60^\circ}$. **2**

13. A ladder makes 45° with the ground and its foot is 4 m from the wall. Find the length of the ladder. **2**

14. A tower casts a 30 m shadow when the sun's elevation is 30°. Find its height. **2**

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that $\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} = \tan\theta + \cot\theta$. **3**

16. Two points are on opposite sides of a 60 m tower, in line with its foot. The angles of elevation of its top from them are 30° and 60°. Find the distance between the points. **3**

17. If $\cos\theta + \sin\theta = \sqrt{2}\cos\theta$, show that $\cos\theta - \sin\theta = \sqrt{2}\sin\theta$. **3**

OR

If $\tan A = n\tan B$ and $\sin A = m\sin B$, prove that $\cos^2 A = \frac{m^2-1}{n^2-1}$. **3**

18. A man on top of a 100 m cliff sees two boats in the same direction at angles of depression 60° and 30°. Find the distance between the boats. **3**

OR

From a point A, the angle of elevation of a tree top is 30°. On walking 30 m towards the tree, it becomes 60°. Find the height of the tree and the distance of the second point from the tree. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that $(\sin A + \cos A)(\tan A + \cot A) = \sec A + \operatorname{cosec} A$. Verify it for $A = 45^\circ$. **5**

OR

If $x = a\sec\theta + b\tan\theta$ and $y = a\tan\theta + b\sec\theta$, prove that $x^2 - y^2 = a^2 - b^2$. **5**

20. **Case study:** From the top of a 48 m building, the angle of elevation of the top of a taller building is 30° and the angle of depression of its foot is 60°. **5**

(i) Find the distance between the buildings.

(ii) Find the extra height above the first building's top.

(iii) Find the height of the taller building.

OR

The angle of elevation of the top of a hill from a point A is 30°. From a point B, 1 km nearer the hill, it is 45°. Find the height of the hill. **5**

END OF QUESTION PAPER — Total: 40 Marks

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